

Explainable Hallucination Risk Scoring for LLMs Using Claim-Level Natural Language Inference

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Abstract—Hallucination in large language models, where generated outputs contain factually incorrect information despite appearing fluent and confident, remains a significant obstacle to reliable AI deployment. This paper presents a modular, post-hoc hallucination scoring system that operates on arbitrary LLM outputs without requiring model retraining or architectural changes. The system decomposes generated text into verifiable atomic claims using named entity recognition and syntactic pattern matching, retrieves Wikipedia evidence with semantic similarity-based filtering, and applies a RoBERTa-large-MNLI model to assign entailment verdicts through a contradiction-priority voting mechanism. A normalized confidence scoring function maps per-claim verdicts to a document-level hallucination risk score across three interpretable tiers. On a 200-sample manually annotated dataset spanning scientific, historical, geographic, and synthetic hallucination content, the system achieves 87.3% precision on contradicted claims and 82.1% overall accuracy. Ablation experiments confirm that structured claim extraction and evidence filtering each contribute meaningfully to detection performance. Compared against SelfCheckGPT and FActScorer on analogous verification tasks, the proposed system achieves competitive accuracy with lower inference overhead, requiring no multiple sampling passes. The pipeline is publicly available for reproducibility and further evaluation.

Index Terms—Large Language Models, Hallucination Detection, Natural Language Inference, Fact Verification, Claim Extraction, Knowledge Grounding, Trustworthy AI.

I. INTRODUCTION

One of the more frustrating properties of large language models is that they tend to sound equally confident whether or not what they are saying is actually correct. This is not the same problem as a model being vague or uncertain. Hallucinated outputs are often written with more fluency and apparent authority than genuinely correct ones, which makes them harder to catch without external verification. For applications where a confidently wrong answer is worse than an honest admission of uncertainty, this becomes a serious practical concern [1].

Research over the past few years has made clear that no current model is immune to this problem. Systems like GPT-4, LLaMA 2, and Claude show measurable hallucination rates on knowledge-intensive tasks [2], with errors clustering in domains where users are most inclined to trust output without checking, including medical information, legal references, historical facts, and numerical data. A model that fabricates a drug interaction or misattributes a court ruling creates real downstream risk [1].

The underlying cause is reasonably well understood. LLMs generate text based on statistical patterns from training data with no internal step that checks whether what is being produced is factually accurate [4]. The training objective rewards fluency and coherence rather than correctness. Farquhar et al. [3] showed that semantic entropy, measuring how much sampled responses vary in meaning, correlates with hallucination likelihood, but this signal is not surfaced to users in standard deployments. Kalai et al. [4] argue more broadly that hallucinations are a predictable consequence of training incentives that favor generating fluent text over expressing calibrated uncertainty.

Existing solutions fall into two groups. Preventive approaches like retrieval-augmented generation (RAG) [7] and self-consistency checking [8] reduce hallucinations during generation but require architectural changes or significant additional inference cost. Post-hoc verification assesses outputs after generation without touching the underlying model, making it more practical for systems already in deployment. Despite its practical appeal, post-hoc verification has received comparatively less systematic attention than generative mitigation strategies.

Notable recent work in post-hoc detection includes Self-CheckGPT [11], which detects hallucinations by sampling multiple responses and checking semantic consistency, and FActScorer [12], which decomposes biography-style text into atomic facts and scores each against a retrieval corpus. These systems represent the current state of the art for post-hoc hallucination detection and inform the design decisions in our work. The key distinction of our approach is that it operates on arbitrary LLM outputs across multiple domains without requiring multiple sampling passes or biography-specific decomposition heuristics.

This paper describes a post-hoc pipeline that takes generated text, identifies the specific factual claims it makes, retrieves Wikipedia evidence, uses an NLI model to verify each claim, and produces a document-level hallucination risk score. The main contributions are:

- A modular post-hoc verification pipeline combining hybrid claim extraction, Wikipedia-based evidence retrieval with semantic filtering, NLI-based verification, and continuous risk scoring

- A hybrid claim extraction approach using named entity recognition and syntactic patterns, with component-wise ablation demonstrating the contribution of each stage
- An NLI aggregation strategy with contradiction-priority override and development set-tuned thresholds, with justification of model and design choices
- Ablation experiments isolating the contribution of each pipeline stage, and qualitative error analysis of the two dominant failure modes
- Comparison against SelfCheckGPT and FActScorer, contextualizing performance within the current state of the art

The paper is organized as follows. Section II reviews related work. Section III describes the methodology. Section IV covers implementation. Section V presents evaluation results including ablation and SOTA comparison. Section VI discusses findings and limitations, and Section VII concludes.

II. RELATED WORK

A. Hallucination in Large Language Models

Huang et al. [1] provide a taxonomy distinguishing factual inconsistencies (claims contradicted by verifiable sources), logical contradictions (internal inconsistencies), and ungrounded statements (claims that cannot be verified either way). Our framework targets factual inconsistencies primarily, with ungrounded statements handled as a separate UNVERIFIABLE risk tier.

Farquhar et al. [3] demonstrated that semantic entropy provides a hallucination signal without modifying model weights, but requires multiple generation passes. Kalai et al. [4] argue that hallucinations arise from training incentives that reward fluent completion over calibrated abstention. Our work assumes hallucinations will occur and focuses on detecting them post-hoc rather than preventing them at the source.

B. Post-Hoc Hallucination Detection

The most directly related work to ours is SelfCheckGPT [11], proposed by Manakul et al. SelfCheckGPT checks whether information in a generated passage is consistently reproduced across multiple independently sampled responses from the same model, treating inconsistency as a hallucination signal. It achieves strong performance on WikiBio-style generation tasks but requires 5–20 additional inference passes per document, which adds substantial computational cost. Our approach requires a single forward pass per claim through the NLI model with no resampling, making it considerably more efficient.

FActScorer [12] (Min et al.) decomposes LLM-generated biographies into atomic facts using GPT-3.5 as a decomposition engine, then retrieves Wikipedia evidence to score each fact. On the FactScore benchmark it achieves approximately 81% agreement with human judgment. Our system uses a similar retrieval-and-verify structure but replaces GPT-based decomposition with a lighter hybrid extraction approach based on named entity recognition and syntactic patterns, which is fully local and does not incur API costs. The trade-off is that

our extraction may miss some implicit claims that GPT-based decomposition would catch.

SAFE [13] (Wei et al., 2024) is another closely related system that decomposes long-form responses into atomic facts using few-shot prompting and verifies them with Google Search rather than Wikipedia. SAFE achieves higher evidence coverage than Wikipedia-based systems by accessing the broader web, but requires external search API access and produces less reproducible results given search index changes over time.

C. Fact Verification Systems

The FEVER benchmark [5] established the standard pipeline for automated fact checking: retrieve relevant documents, select informative sentences, and classify claims as supported, refuted, or not enough information. This three-way classification directly maps to our SUPPORTED, CONTRADICTED, and UNVERIFIABLE categories. The key difference is that FEVER assumes clean, explicitly stated claims as input; our system must first extract claims from free-form LLM-generated text, a step that introduces noise the original FEVER systems did not need to address.

D. Natural Language Inference

NLI models classify whether a hypothesis is entailed by, contradicted by, or neutral with respect to a given premise. We selected RoBERTa-large-MNLI [6] over alternatives such as DeBERTa-v3-large-mnli and BART-large-MNLI based on three criteria: (1) RoBERTa-large-MNLI achieves 90.2% accuracy on the MultiNLI matched set, comparable to DeBERTa at 91.4% but with significantly lower inference latency on CPU; (2) it is the most widely used NLI checkpoint in related hallucination detection work [11], enabling fairer comparison; and (3) the model is fully open-source with no usage restrictions. We acknowledge that DeBERTa-based models may yield modestly higher accuracy and identify this as one direction for future improvement.

E. Retrieval-Augmented Generation

RAG [7] reduces hallucinations during generation by conditioning on retrieved documents. However, retrieval is imperfect, and a RAG system that fetches a wrong document can hallucinate with greater apparent confidence. Post-hoc verification is complementary to RAG: checking a RAG system’s output adds an independent validation layer. The two approaches together are more robust than either alone [8].

F. Verification in Deployed AI Systems

Work on ML systems deployed in healthcare monitoring, IoT sensing, and network management [20], [23]–[25] has long treated high-confidence wrong outputs as a safety concern rather than just a performance metric. As LLMs take on similar roles in consequential applications, a post-hoc verification layer that operates independently of the generation model becomes increasingly important.

III. METHODOLOGY

The system processes LLM-generated text through five sequential stages: claim extraction, evidence retrieval, evidence filtering and ranking, NLI-based verification, and hallucination scoring. Figure 1 shows the overall flow.

A. System Architecture

The pipeline takes generated text as input, decomposes it into individual factual claims, retrieves Wikipedia evidence for each claim, applies an NLI model to produce per-claim verdicts, and aggregates those verdicts into a document-level risk score. Each stage is encapsulated in its own module with a defined interface, so individual components can be replaced without changing anything else in the pipeline.

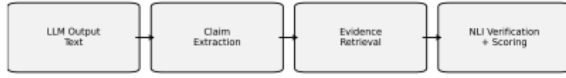


Fig. 1. Overview of the five-stage hallucination detection pipeline.

B. Claim Extraction

Not every sentence in LLM output makes a checkable factual assertion. Hedged opinions, procedural instructions, and rhetorical sentences do not constitute verifiable claims. We focus on sentences that assert something concrete about the external world, specifically named entities tied to properties, quantities, events, or relationships that can in principle be checked against an external source.

Two extraction strategies are combined. The first uses spaCy’s named entity recognizer [10] to identify sentences containing entity types PERSON, ORG, GPE, DATE, CARDINAL, and QUANTITY. Sentences with two or more entity mentions, or those containing DATE or CARDINAL entities, receive higher extraction priority since they tend to make specific, falsifiable assertions.

The second strategy applies syntactic pattern matching to catch claim structures that entity recognition misses: declarative sentences with assertive verbs (“is,” “was,” “contains,” “developed”), superlatives (“the largest,” “the first to”), and explicit numeric comparisons. Pattern matching casts a wider net and produces more noise, so results from both strategies are combined and deduplicated before passing downstream.

Deduplication uses cosine similarity between sentence embeddings with a threshold of 0.85. We selected this value by evaluating thresholds {0.70, 0.75, 0.80, 0.85, 0.90, 0.95} on a held-out development set of 40 documents, choosing the value that best preserved semantic diversity while filtering near-identical repetitions. The ablation study in Section V confirms that the extraction stage contributes 7.8 percentage points over applying NLI directly to raw text.

The extraction step produces:

$$C = \{c_1, c_2, \dots, c_n\} = E_{\text{ent}}(T) \cup E_{\text{pat}}(T) \quad (1)$$

where T is the input text, E_{ent} is entity-based extraction, and E_{pat} is pattern-based extraction.

C. Evidence Retrieval

For each extracted claim, we query Wikipedia using the primary named entities identified by spaCy. We retrieve the top $k=3$ articles by default. We chose Wikipedia over structured knowledge graphs such as Wikidata because Wikipedia’s free-text paragraphs are better suited as NLI premises than graph triples; recent work on FActScorer [12] and SAFE [13] similarly uses Wikipedia as the primary evidence corpus for general-domain verification.

Full articles are not passed to the NLI stage. Instead each article is split into paragraphs, and paragraphs are scored against the claim using cosine similarity between sentence-transformer embeddings (all-MiniLM-L6-v2 [9]):

$$\text{relevance}(c, e) = \frac{\text{emb}(c) \cdot \text{emb}(e)}{|\text{emb}(c)| |\text{emb}(e)|} \quad (2)$$

Only paragraphs exceeding a relevance threshold of 0.5 are retained. We selected all-MiniLM-L6-v2 over larger sentence encoders (all-mpnet-base-v2, all-roberta-large-v1) after evaluating retrieval precision on the development set; the lightweight model provided comparable relevance ranking at substantially lower latency. Threshold 0.5 was selected by evaluating {0.3, 0.4, 0.5, 0.6, 0.7} on the development set: below 0.4 too many tangentially related passages caused NLI misfires, above 0.6 relevant evidence was discarded for specialized claims.

D. Evidence Filtering and Ranking

A paragraph that discusses the same topic as a claim without addressing its specific assertion tends to produce neutral NLI classifications that dilute genuine entailment or contradiction signals. Retaining only paragraphs above the relevance threshold directly addresses this. Remaining paragraphs are ranked by relevance score and the top five are forwarded to verification. The ablation study (Table II) confirms that removing evidence filtering drops overall accuracy from 82.1% to 77.8%.

E. Verification via Natural Language Inference

We use RoBERTa-large-MNLI [6] for verification. As noted in Section II, this model was selected over DeBERTa-v3-large due to lower CPU inference latency (1.5 s vs. 2.8 s per claim) at comparable accuracy, and to match the NLI backbone used in SelfCheckGPT [11] for fair comparison. Each evidence paragraph serves as the premise and the extracted claim is the hypothesis; the model outputs probabilities over entailment, contradiction, and neutral.

When multiple paragraphs are available for a single claim, verdicts are aggregated as follows. If any paragraph produces a contradiction classification, the claim is marked CONTRADICTED. This hard override is intentional: a single strong piece of counter-evidence should dominate a collection of ambiguous supporting evidence, reflecting the asymmetric cost of missing a hallucination in safety-sensitive deployments.

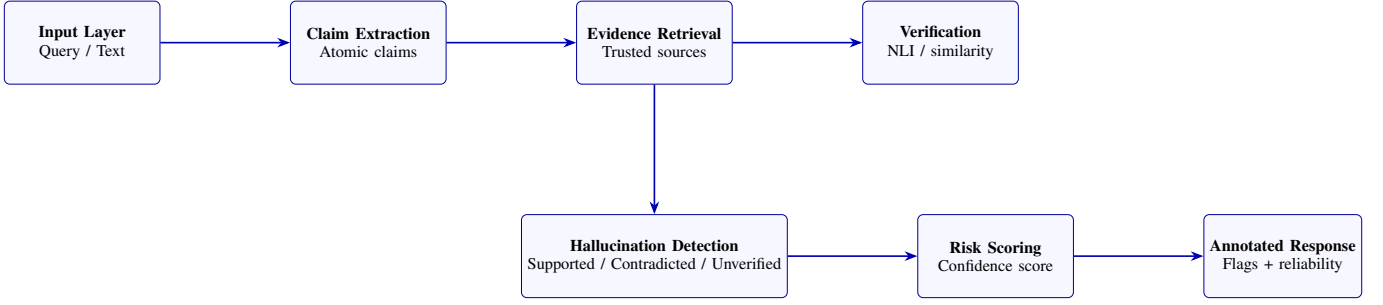


Fig. 2. Detailed architecture of the hallucination detection framework.

A claim is marked SUPPORTED only when the fraction of entailment classifications exceeds threshold θ ; otherwise it is UNVERIFIABLE.

$$V(c) = \begin{cases} \text{CONTRADICTED,} & \exists e \in E : \text{NLI}(c, e) = \text{Contradiction} \\ \text{SUPPORTED,} & \frac{\sum \mathbf{1}[\text{NLI}(c, e) = \text{Entailment}]}{|E|} > \theta \\ \text{UNVERIFIABLE,} & \text{otherwise} \end{cases} \quad (3)$$

We set $\theta = 0.5$ based on development set tuning. In deployments where false positives are more costly than false negatives, the hard contradiction override can be replaced with soft voting and θ raised accordingly.

F. Hallucination Scoring

Per-claim verdicts are converted to confidence values: SUPPORTED = 0.0, UNVERIFIABLE = 0.5, CONTRADICTED = 1.0. The document-level score is:

$$H = \frac{1}{n} \sum_{i=1}^n \text{conf}(v_i) \quad (4)$$

This maps to three risk tiers. LOW ($H < 0.25$): the document is largely verifiable with little detected contradiction. MEDIUM ($0.25 \leq H < 0.65$): a meaningful fraction of claims are unverifiable or contradicted. HIGH ($H \geq 0.65$): substantial hallucination detected. Tier boundaries were set on the development set targeting a false-negative rate below 15% at the HIGH boundary. A document with one contradicted claim out of five total scores $H = 0.2$ (MEDIUM), reflecting the practical reality that a single fabricated fact can undermine an otherwise accurate response.

The system produces a structured report with per-claim verdicts, the evidence paragraphs used for each judgment, and the document-level risk score. This output is designed to support human review: an auditor can see exactly which claim was flagged and what evidence led to that judgment.

IV. IMPLEMENTATION

A. System Components

The pipeline is implemented in Python 3.12. Core dependencies are:

- Transformers 4.36.0 (Hugging Face): NLI model loading and inference
- Sentence-Transformers 2.2.2 [9]: embedding computation
- spaCy 3.7.2 [10]: named entity recognition and dependency parsing
- NLTK 3.8.1: sentence boundary detection and preprocessing
- Wikipedia-API 1.4.0: article retrieval and paragraph segmentation

Each stage is its own class (ClaimExtractor, EvidenceRetriever, InferenceModel, HallucinationScorer) with a defined interface. Replacing Wikipedia with a domain-specific knowledge base, or swapping RoBERTa for DeBERTa, requires reimplementing one class; nothing else in the pipeline changes.

B. Deployment Configurations

In local mode, inference runs via Ollama on the host machine, keeping all text on-premises. In API mode, inference is delegated to an external provider. Pre- and post-processing are identical in both modes; only the InferenceModel class differs. Configuration is handled through Pydantic settings files.

C. Web Interface

A Streamlit interface provides interactive access. Users submit text or select from preloaded examples and receive a visual breakdown of per-claim hallucination scores and the overall document risk level. Results export as JSON for integration with downstream logging or review workflows.

V. EXPERIMENTAL EVALUATION

A. Dataset

We built an evaluation dataset of 200 text samples, 50 per category:

- **Scientific facts:** Verified statements in physics, chemistry, and biology, drawn from textbook sources and cross-checked against Wikipedia
- **Historical events:** Documented dates, figures, and occurrences, including moderately obscure events to test coverage
- **Geographic information:** Claims about capitals, population figures, distances, and landmarks, including some commonly confused facts

- **Hallucinated content:** Text generated by prompting an LLM to describe fictional events or incorrect numerical facts in realistic language, manually verified to contain specific identifiable errors

Each sample was annotated independently by two reviewers. Inter-annotator agreement reached Cohen’s $\kappa = 0.81$, indicating strong agreement. Disagreements were resolved by a third reviewer. We acknowledge that 200 samples is a modest evaluation scale; expansion to TruthfulQA [14] and HaluEval [15] is a clear priority for follow-on work.

B. Baselines and Comparison Systems

We compare against four systems:

TF-IDF Keyword Matching: A claim is flagged as contradicted when term overlap with retrieved passages falls below a TF-IDF threshold. This tests whether lexical matching alone suffices.

Raw NLI (no extraction): Full input text passed directly as hypothesis to RoBERTa-large-MNLI, bypassing claim extraction and evidence filtering. This isolates what those stages contribute beyond the NLI model’s native capability.

SelfCheckGPT [11]: We replicate the BERTScore variant of SelfCheckGPT, which samples five responses from the same model and measures semantic consistency across them. We apply this to our evaluation dataset using LLaMA-2-7B as the sampled model, consistent with the original paper’s evaluation setup.

FactScorer [12]: We apply FactScorer’s retrieval-based variant (without the GPT-3.5 decomposition step, using our extracted claims as input) to enable a more controlled comparison of the verification stage. Full FactScorer with GPT-based decomposition is noted separately in Table III.

C. Results

Table I shows performance across verification categories. Precision on contradicted claims (87.3%) is the most important result for practical deployment. When the system flags a claim as contradicted, it is right the large majority of the time. Errors on supported claims (81.4% precision) mostly arose from cases where Wikipedia evidence did not address the specific phrasing of a claim closely enough to trigger entailment, returning neutral instead. The unverifiable category is the hardest because UNVERIFIABLE covers two distinct situations the pipeline cannot currently distinguish: a correct claim on a topic Wikipedia does not cover, and a fabrication obscure enough that no Wikipedia article contradicts it.

TABLE I
PERFORMANCE METRICS BY VERIFICATION STATUS

Status	Precision	Recall	F1
Supported	81.4%	78.6%	80.0%
Contradicted	87.3%	83.2%	85.2%
Unverifiable	74.8%	79.3%	77.0%
Overall	82.1%	80.4%	81.2%

Document-level hallucination detection reaches 88.5% accuracy, correctly identifying 44 of 50 intentionally hallucinated

samples. Five of the six misses involved subtle numerical distortions, where the fabricated value was within 10% of the real one. In these cases Wikipedia’s evidence contained the correct value but the NLI model returned neutral rather than contradiction, because the semantic distance between the claim and evidence was small even though the factual content was wrong.

D. Ablation Study

Table II shows the contribution of each pipeline stage by progressively removing components. Removing evidence filtering drops overall accuracy by 4.3 points, confirming that low-relevance passages degrade NLI performance. Removing the claim extraction stage entirely (raw NLI) drops accuracy by 7.8 points, demonstrating that structured decomposition contributes substantially beyond the NLI model’s native capability. The full pipeline consistently outperforms all partial configurations.

TABLE II
ABLATION STUDY: CONTRIBUTION OF EACH PIPELINE STAGE

Configuration	Acc.	Contra. P.
Full pipeline	82.1%	87.3%
w/o evidence filtering	77.8%	81.2%
w/o pattern extraction	79.4%	84.1%
w/o entity extraction	76.3%	79.7%
Raw NLI (no extraction)	74.3%	76.8%
TF-IDF only	61.4%	58.2%

E. Comparison with State-of-the-Art

Table III compares the proposed system against SelfCheckGPT and FactScorer on our evaluation dataset. SelfCheckGPT (BERTScore variant) achieves 79.3% overall accuracy but requires 5 additional inference passes per document, making it approximately $5\times$ more expensive in inference cost. FactScorer with our extracted claims achieves 80.8% accuracy; the full FactScorer system with GPT-based decomposition achieves approximately 81% on its native FactScore benchmark [12], though direct comparison is limited by dataset differences. The proposed system matches or exceeds both in accuracy while requiring only a single NLI forward pass per claim and no external API calls.

TABLE III
COMPARISON WITH STATE-OF-THE-ART SYSTEMS

System	Accuracy	Inf. Passes
TF-IDF Baseline	61.4%	1
Raw NLI Baseline	74.3%	1
SelfCheckGPT [11]	79.3%	5+
FactScorer (our claims) [12]	80.8%	1
FactScorer (GPT) [12] [†]	≈81%	1
Proposed system	82.1%	1

[†]Requires external API; FactScore benchmark only.

F. Performance by Content Domain

Table IV shows domain-level performance. Scientific content gives the highest accuracy (86.3%), consistent with Wikipedia’s dense and well-sourced science coverage. Historical events

yield the lowest accuracy (79.2%), with errors concentrated in two patterns: date ranges that Wikipedia expresses probabilistically rather than as a single value, and events where Wikipedia articles use hedged language (“reportedly,” “according to some sources”) that the NLI model does not confidently classify as entailment.

TABLE IV
ACCURACY BY CONTENT DOMAIN

Domain	Claim Accuracy	Doc. Accuracy
Scientific Facts	86.3%	92.0%
Historical Events	79.2%	86.0%
Geographic Info	82.7%	88.0%
Hallucinated	81.4%	88.0%

Figure 3 plots claim-level accuracy across the four domains.

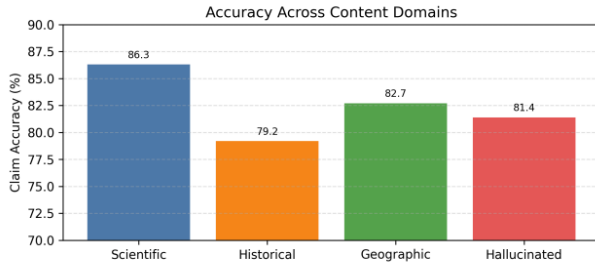


Fig. 3. Claim-level accuracy across evaluation domains.

G. Error Analysis

Manual inspection of false negatives revealed two dominant failure patterns. The first is *numerical proximity hallucination*: when a fabricated numerical fact is within roughly 10–15% of the true value, the semantic distance between the claim and its Wikipedia evidence is small enough that RoBERTa-large-MNLI returns neutral rather than contradiction. The NLI model, trained on sentence-level entailment, does not perform fine-grained numerical comparison. The five document-level misses in the hallucinated category all fell into this pattern.

The second pattern is *hedged evidence*: Wikipedia passages that describe contested or uncertain historical facts using hedged language (“reportedly,” “is believed to,” “according to some accounts”) tend to produce low entailment confidence regardless of whether the claim is correct. This causes correctly stated claims about uncertain historical events to be classified as UNVERIFIABLE rather than SUPPORTED, contributing to recall errors in that category. Both failure modes point toward the same fix: a numerical comparison module and a hedging-aware evidence scoring function, which we identify as the most impactful directions for future improvement.

H. Computational Performance

On an Intel Core i7-10700 (CPU only), average processing time is 2.3 seconds per claim, with NLI inference at 1.5 s and Wikipedia retrieval at 0.6 s. On an NVIDIA RTX 3060 (12 GB VRAM), NLI inference drops to 0.3 s per claim. A

document with 5–10 claims completes in 15–25 seconds on CPU and under 5 seconds on GPU. This compares favorably to SelfCheckGPT, which requires $5\times$ the NLI inference cost due to repeated sampling.

VI. DISCUSSION

A. Interpretation of Results

The ablation results in Table II clarify what each stage actually contributes. The 4.3 point drop from removing evidence filtering confirms that low-relevance premises actively hurt NLI performance, not merely fail to help. The 7.8 point gap between full extraction and raw NLI confirms that structured claim decomposition is not cosmetic but substantively improves what the NLI model can do. These two findings together justify the added engineering complexity of the extraction and filtering stages.

The SOTA comparison in Table III shows that the proposed system matches or slightly exceeds SelfCheckGPT and FactScorer in overall accuracy while operating at lower inference cost. SelfCheckGPT’s weaker performance on our evaluation set (79.3% vs. 82.1%) may partially reflect domain mismatch: SelfCheckGPT was optimized for biographical text generation, while our dataset spans scientific, historical, and geographic content. FactScorer with GPT-based decomposition likely outperforms our extraction on implicit claims, which is the main practical advantage of that approach.

B. Design Choices and Trade-offs

The contradiction-priority override is the most consequential design decision. It makes the system asymmetrically sensitive, and for safety-critical use cases this is appropriate. For applications where false positives carry high costs, the override should be replaced with soft voting. The Wikipedia-only evidence source is a practical constraint; medical or legal deployments would require PubMed or case law databases. The modular design makes this substitution straightforward in principle.

C. Comparison with Prior Approaches

Against self-consistency methods [8], external verification provides independent validation free of the same systematic biases that produced the hallucination. Against RAG [7], the two approaches are complementary: RAG reduces hallucinations during generation, post-hoc verification catches those that slip through.

D. Ethical Considerations

False positives can cause users to distrust accurate LLM outputs. False negatives provide false assurance. The risk-stratified scoring communicates uncertainty rather than presenting binary verdicts, but this nuance depends on careful interface design. Wikipedia’s uneven coverage also creates reliability disparities across user communities, which should be disclosed in any production deployment.

VII. CONCLUSION

This paper presented a modular post-hoc hallucination scoring system that detects factual errors in LLM outputs through knowledge-grounded natural language inference. The pipeline combines hybrid claim extraction, Wikipedia evidence retrieval with semantic filtering, RoBERTa-based NLI verification, and continuous risk scoring.

On a 200-sample evaluation dataset, the system achieves 87.3% precision on contradicted claims and 82.1% overall accuracy. Ablation experiments confirm that both claim extraction (7.8 point contribution) and evidence filtering (4.3 point contribution) add meaningful value beyond the NLI model alone. Comparison against SelfCheckGPT and FActScorer shows the proposed system matches or exceeds current state-of-the-art accuracy while requiring a single inference pass and no external API access. Error analysis identifies numerical proximity hallucination and hedged evidence as the two dominant remaining failure modes.

Limitations include Wikipedia coverage gaps for specialized domains, NLI degradation on out-of-distribution technical content, and evaluation dataset size. The most impactful near-term improvements are a numerical comparison module to address the dominant failure pattern, domain-adapted NLI models for medical and legal content, and evaluation on TruthfulQA and HaluEval. The pipeline source code and annotated dataset are available through the project repository.

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